

Compact

center - stable

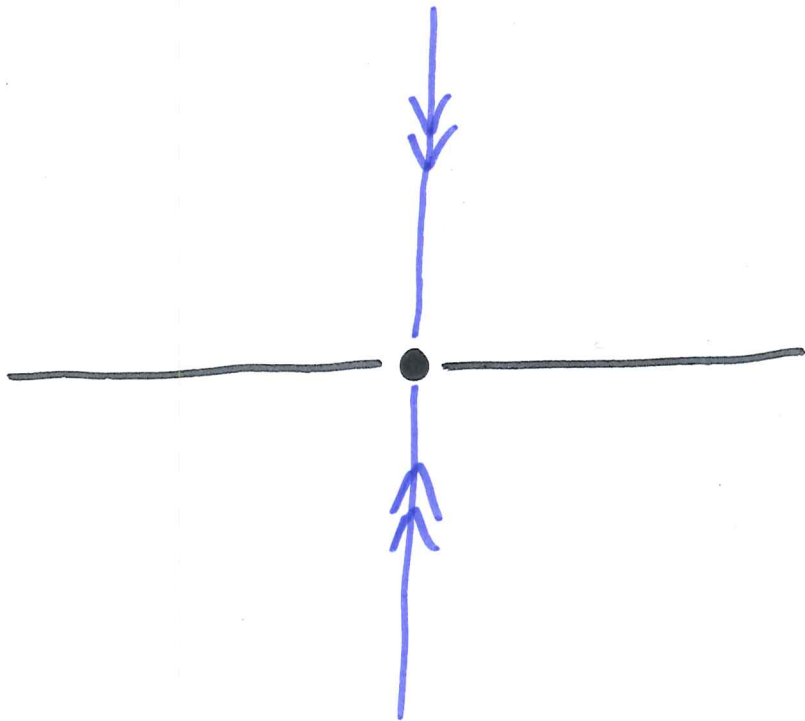
submanifolds.

Andy Hammerlindl

Consider $f(x, y) = (\frac{1}{2}x, \frac{1}{4}y)$ on $\mathbb{R}^2$.

The origin has a well-defined strong stable manifold.

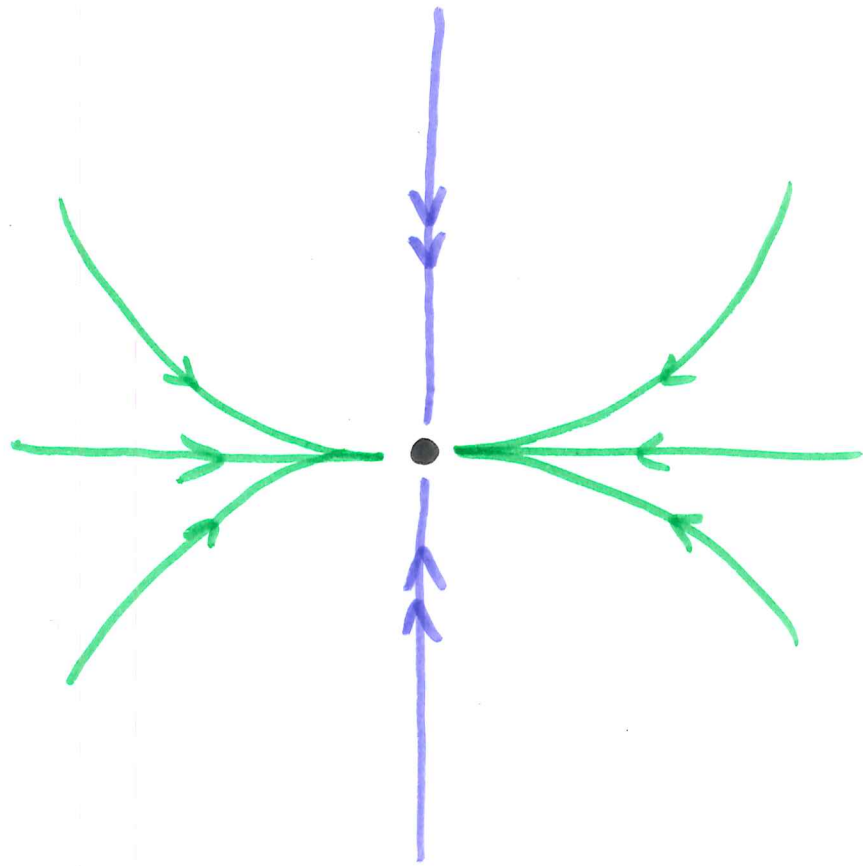
$$\omega^s(0) = \left\{ p \in \mathbb{R}^2 : \lim_{n \rightarrow \infty} \frac{d(f^n(p), 0)}{\lambda^n} = 0 \right\}$$



$$\text{where } \frac{1}{4} < \lambda < \frac{1}{2}.$$

$$f(x, y) = \left(\frac{x}{2}, \frac{y}{4} \right)$$

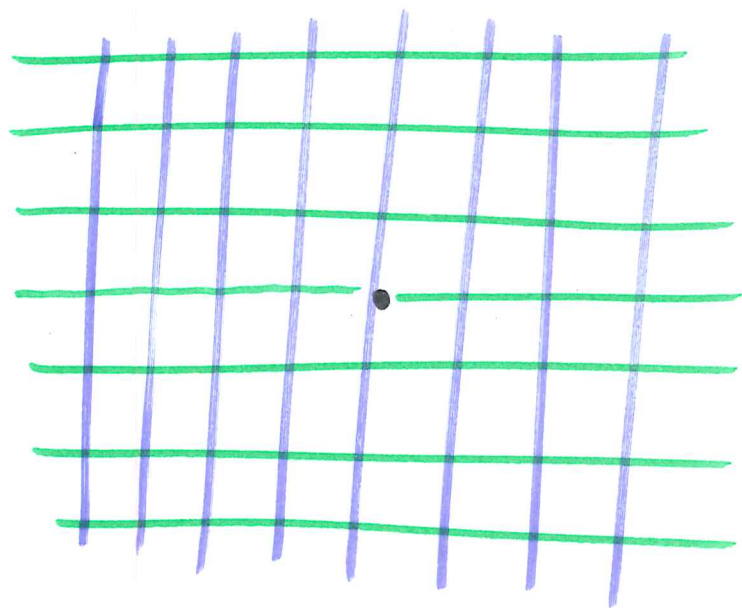
The center or slow manifold is
not well-defined.



For any $c \in \mathbb{R}$,
consider the curve

$$y = cx^2.$$

We want a well-defined **center** direction everywhere in the phase space.



Look at systems
 M is compact.

$f: M \rightarrow M$ where

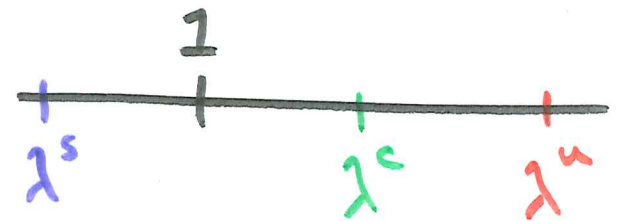
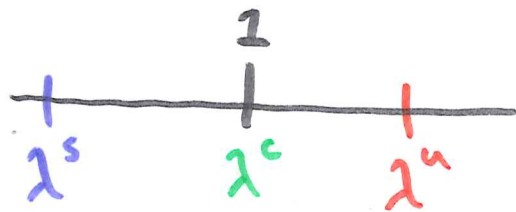
Consider a 3×3 matrix A with integer entries,
 $\det(A) = 1$, and eigs $\lambda^s < \lambda^c < \lambda^u$

Ex

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

OR

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$



This defines a map $f: \mathbb{R}^3 / \mathbb{Z}^3 \rightarrow \mathbb{R}^3 / \mathbb{Z}^3$.

3x3 matrix

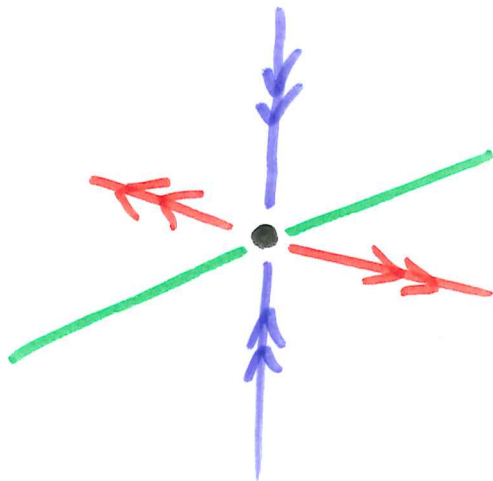


$$f : \mathbb{R}^3 / \mathbb{Z}^3 \hookrightarrow$$

Eigenvalues

$$\lambda^s < \lambda^c < \lambda^u$$

Then, the eigenspaces yield well-defined
stable, center, and unstable
directions at every point.



Def A diffeomorphism $f: M \rightarrow M$ is partially hyperbolic if there is a Df -invariant splitting of the tangent bundle

$$TM = E^s \oplus E^c \oplus E^u$$

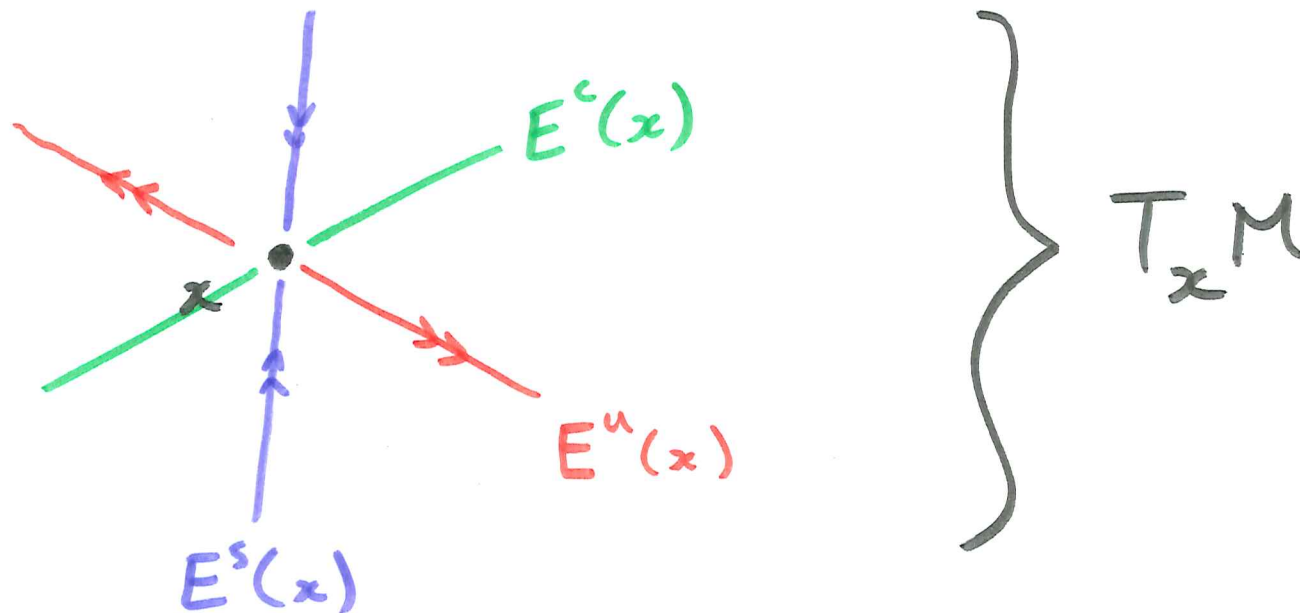
contracted by Df

at most mild contraction or expansion

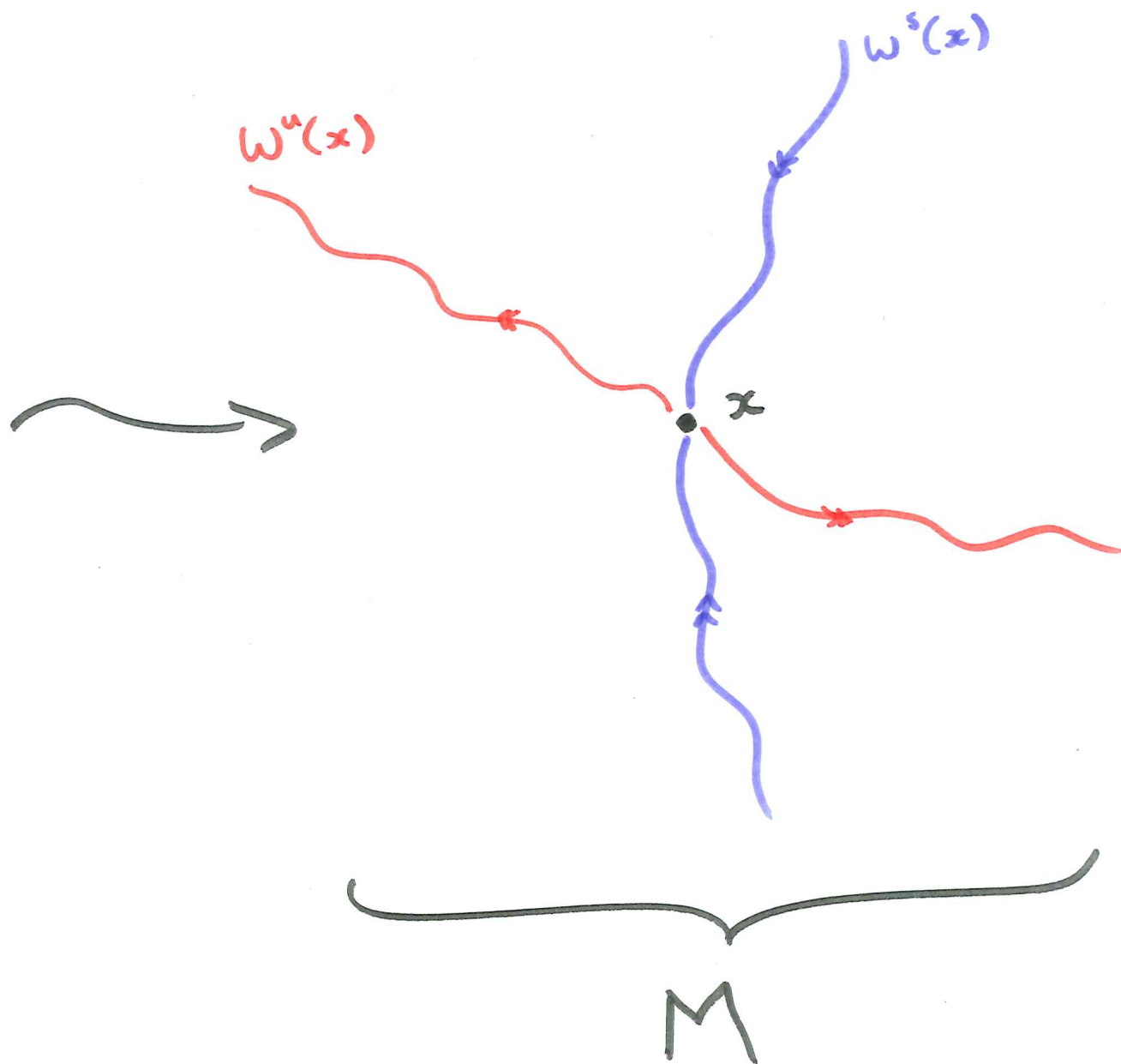
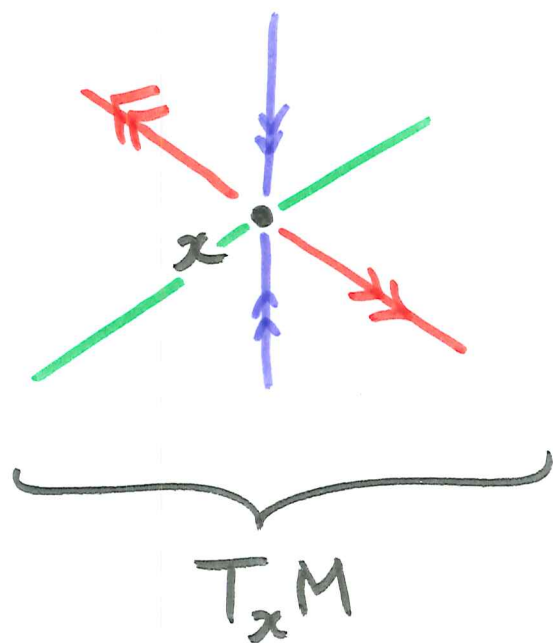
expanded by Df

$$TM = E^s \oplus E^c \oplus E^u$$

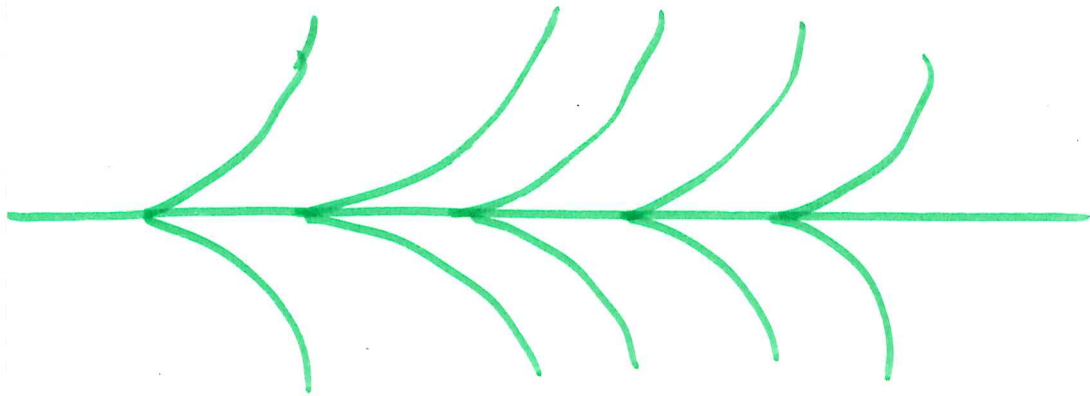
For now, assume $\dim M = 3$ and
each of E^s , E^c , E^u is one-dim'l.



The bundles E^s and E^u are uniquely integrable.



E^c is only Hölder continuous, not C^1 in general.



(Think of
 $\dot{y} = \sqrt[3]{y}$)

For a long time, it was an open question if $\dim E^c = 1$ implies that

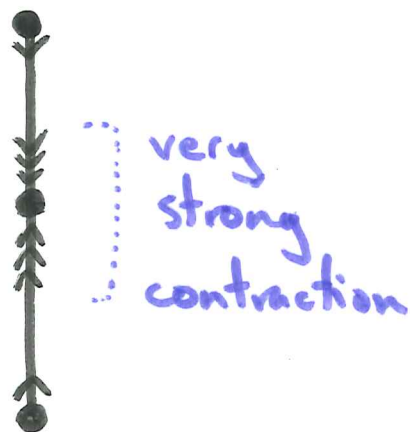
E^c is uniquely integrable.

Counterexample by F. Rodriguez-Hertz, J. Rodriguez-Hertz,
and R. Ures (2016).

There is a partially hyperbolic
diffeomorphism on $\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3$
such that E^c is not uniquely
integrable.

Idea of construction:

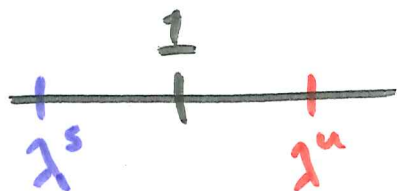
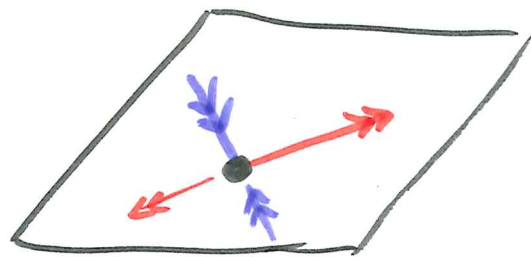
$$g: S^1 \hookrightarrow$$



$$A: \mathbb{R}^2 / \mathbb{Z}^2 \hookrightarrow$$

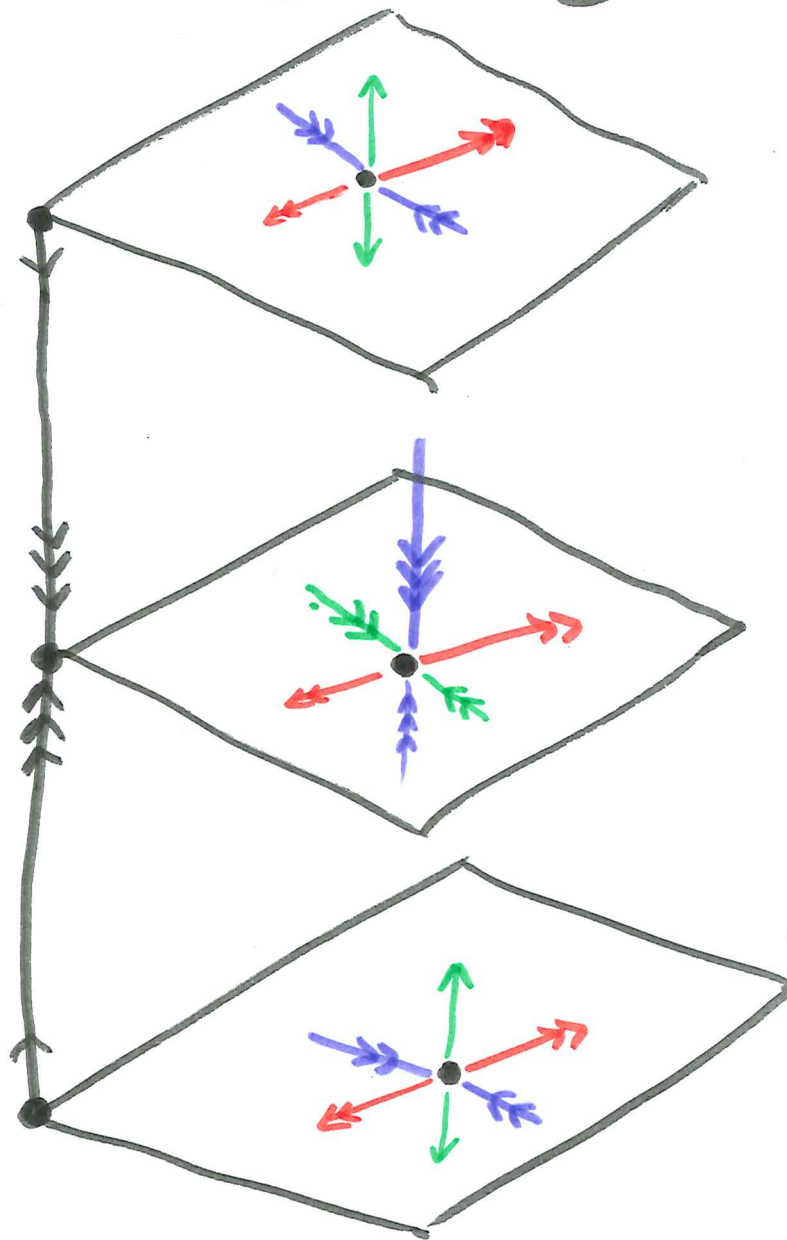
given by
 $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

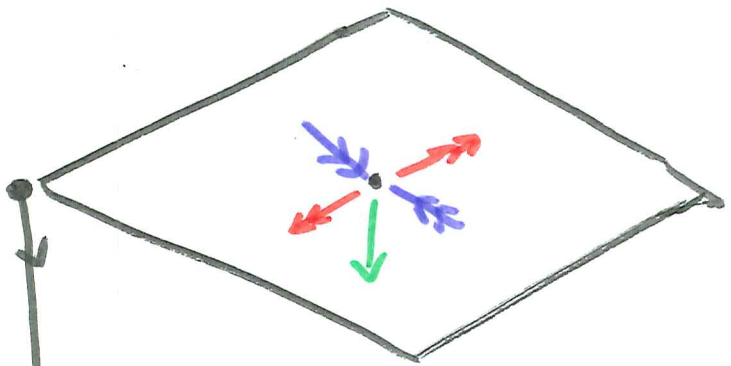
"cat map"



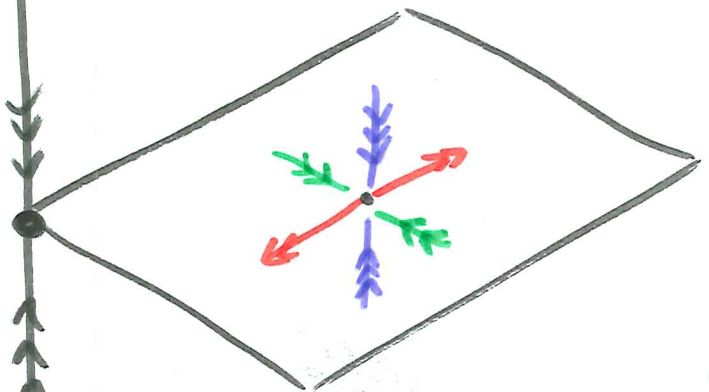
Consider the product

$$g \times A: S^1 \times \mathbb{T}^2 \hookrightarrow$$

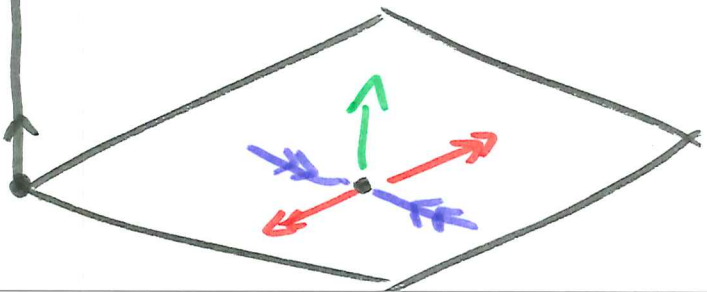




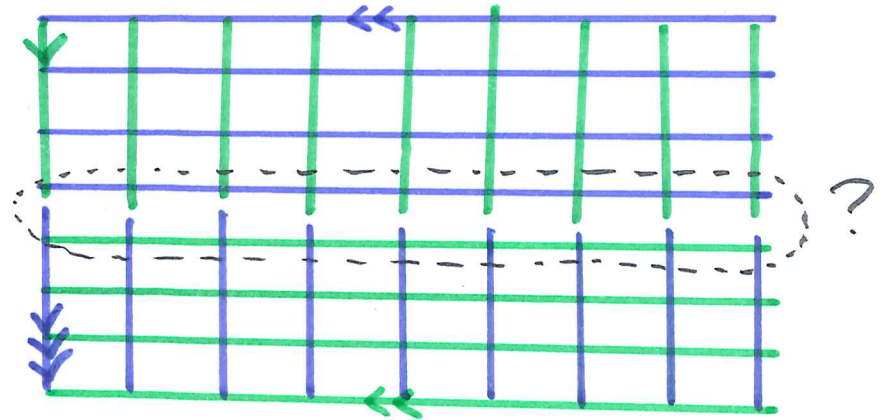
add
Shear



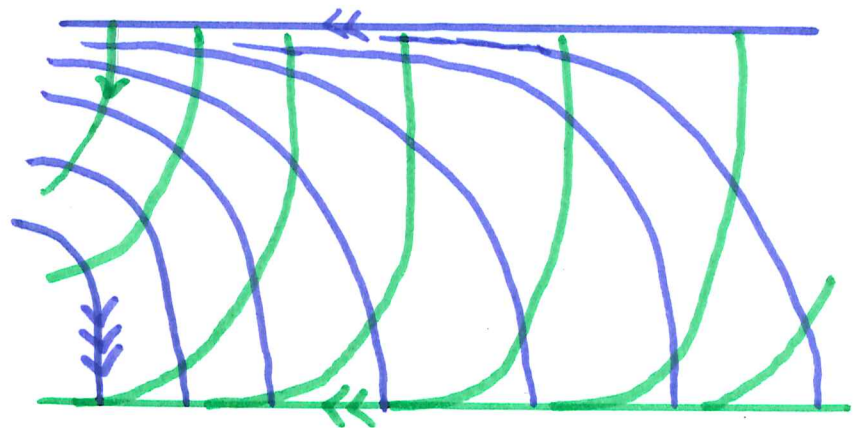
add
shear



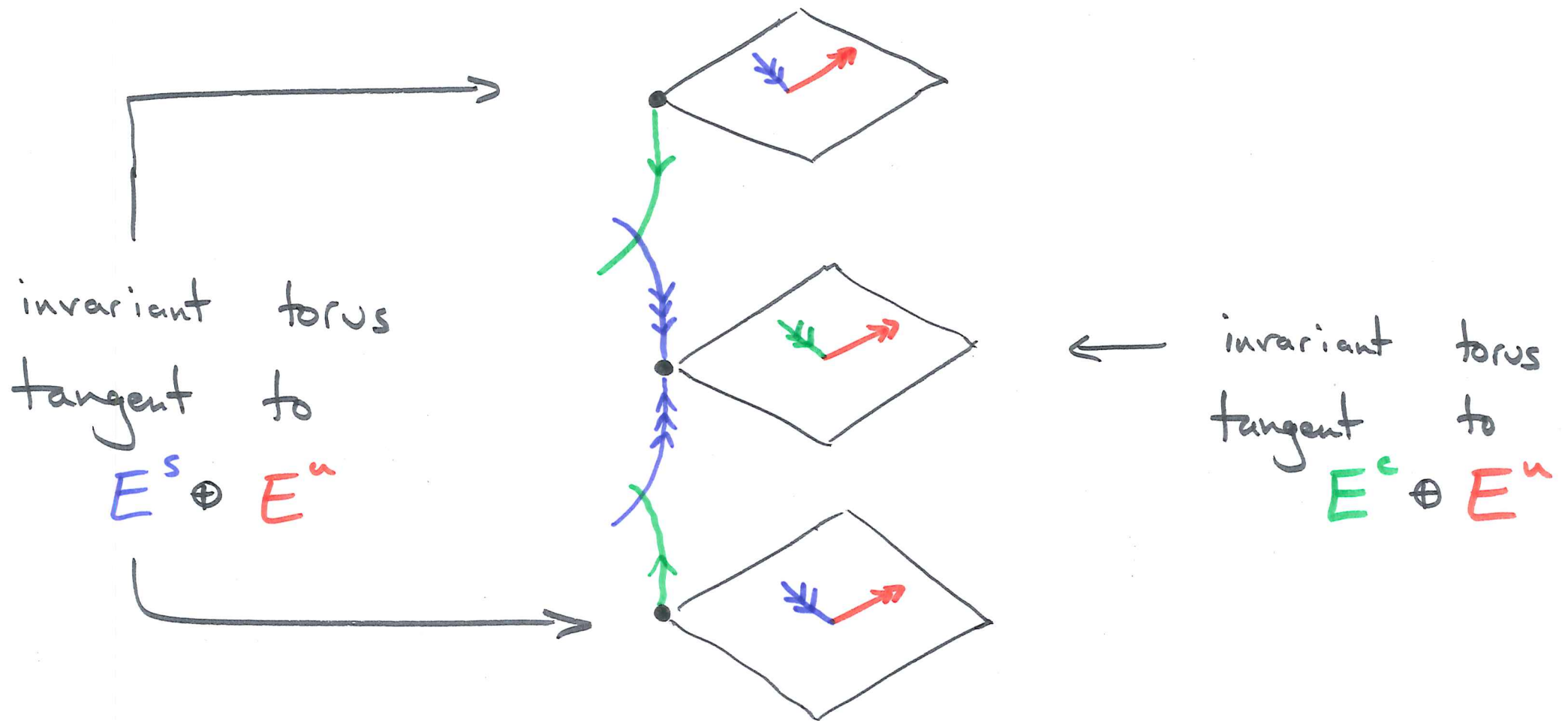
before shear



after shear



$f: M \rightarrow M$ with globally defined splitting $TM = E^s \oplus E^c \oplus E^u$

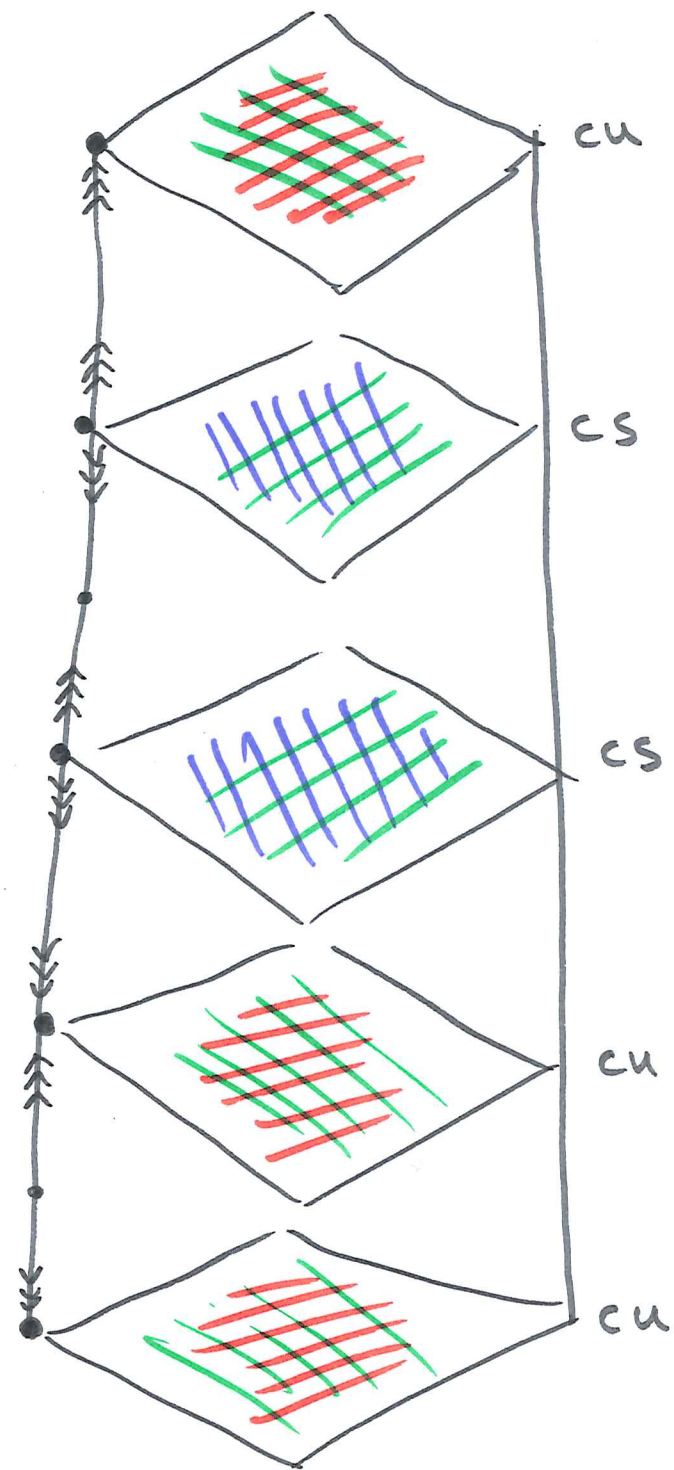


Question 1 In a 3 dim'd partially hyperbolic system, if E^c is not uniquely integrable, must there be an invariant torus tangent to $E^c \oplus E^u$ or $E^c \oplus E^s$?

Question 2 What kinds of systems have compact submanifolds tangent to $E^c \oplus E^u$ or $E^c \oplus E^s$?

R. Potrie and I showed that
 if a 3-diml partially hyperbolic
 system has at least one torus
 tangent to $E^c \oplus E^u$ or $E^c \oplus E^s$,

then it is "basically" a finite
 number of copies of this
 example glued together.



In these 3 dim'l systems, all cs and cu-tori are disjoint.

In higher dimensions, there is more flexibility in creating examples.

Question 3

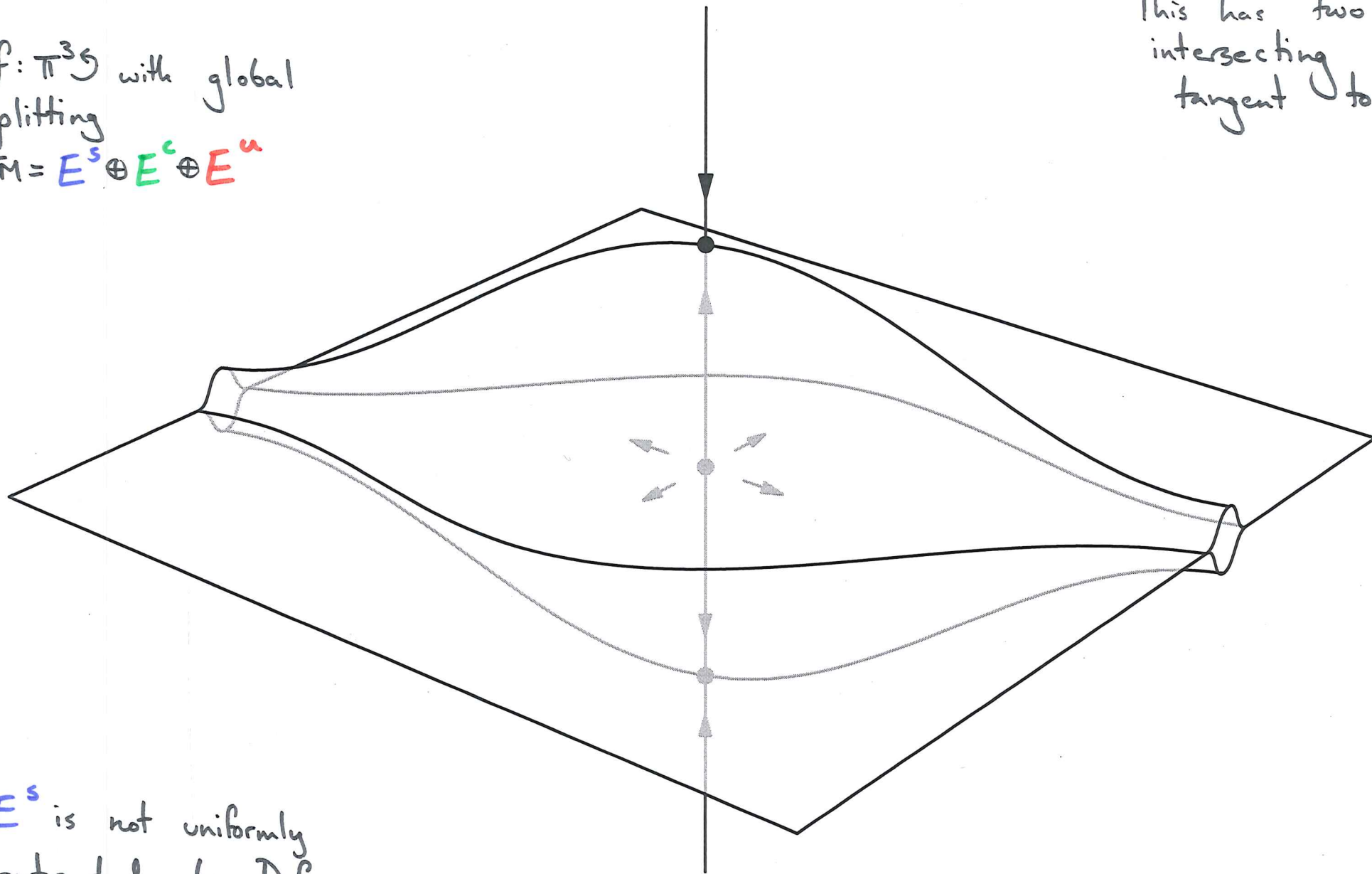
In higher dimensions, do compact submanifolds tangent to $E^c \oplus E^u$ have to be disjoint?

Almost counterexample in dimension 3.

$f: \pi^3 \mathbb{S}$ with global splitting

$$TM = E^s \oplus E^c \oplus E^u$$

This has two intersecting tori tangent to $E^c \oplus E^u$



E^s is not uniformly contracted by Df

What about

dimension 4

or higher?

Thm Let $g: M \rightarrow M$ be any* 3d partially hyperbolic diffeo.

Then there is a 4d partially hyperbolic diffeo $f: M \times S^1 \rightarrow M \times S^1$ such that

- (1) $M \times \{0\}$ is an invariant submanifold tangent to $E_f^c \oplus E_f^u$
- (2) $f(x, 0) = (g(x), 0)$ for all $x \in M$,
- (3) f does not have a centre foliation.

*preserving orientation of E^s

There are many, many 3-manifolds that support partially hyperbolic diffeos.

These produce many many examples in 4-dim without center foliations.

Many of these 4-d examples can
be made absolutely partially hyperbolic

(In 3-d, they cannot.)

Extensions to even higher dimensions.

Theorems available on request.

